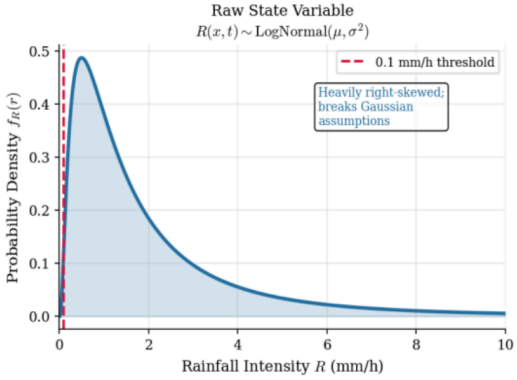
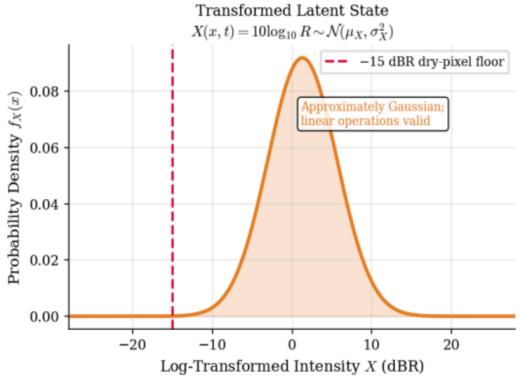
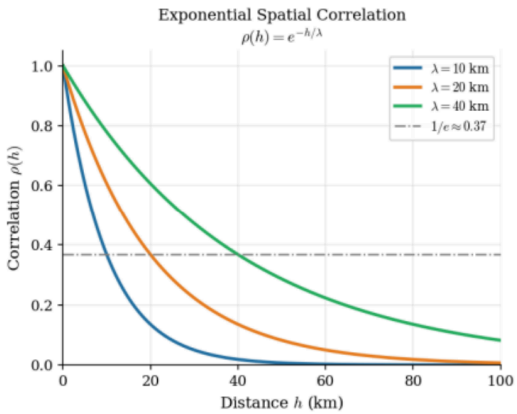
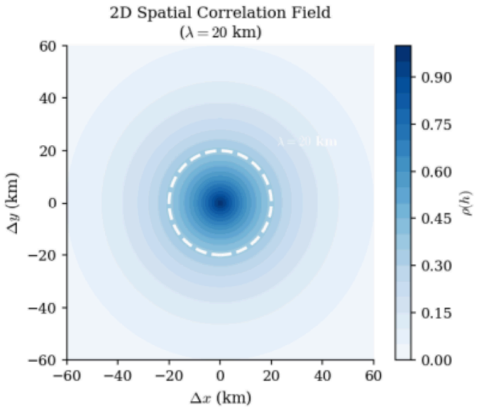
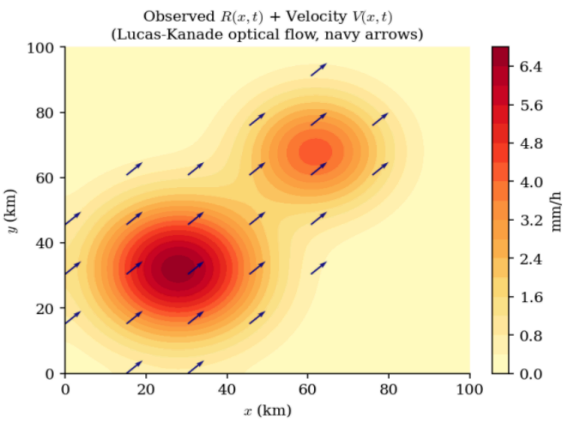
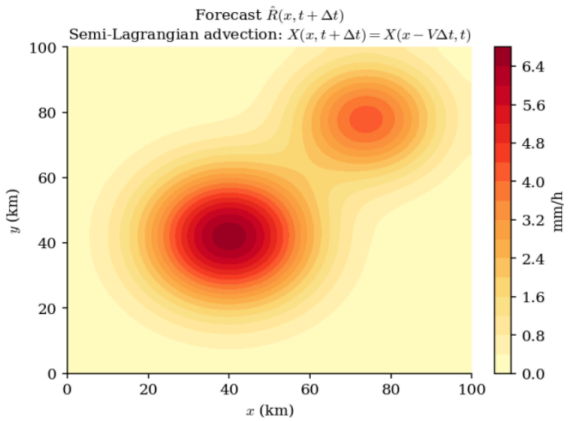

Raw State Variable
$R(x,t) \sim \mathrm{LogNormal}(\mu, \sigma^2)$
0.1 mm/h threshold
Heavily right-skewed; breaks Gaussian assumptions
Probability Density $f_R(r)$
Rainfall Intensity R (mm/h)

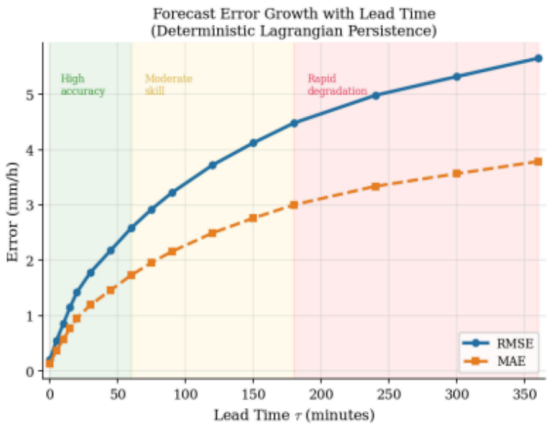

Transformed Latent State
$X(x,t) = 10\log_{10} R \sim \mathcal{N}(\mu_X, \sigma_X^2)$
−15 dBR dry-pixel floor
Approximately Gaussian; linear operations valid
Probability Density $f_X(x)$
Log-Transformed Intensity X (dBR)

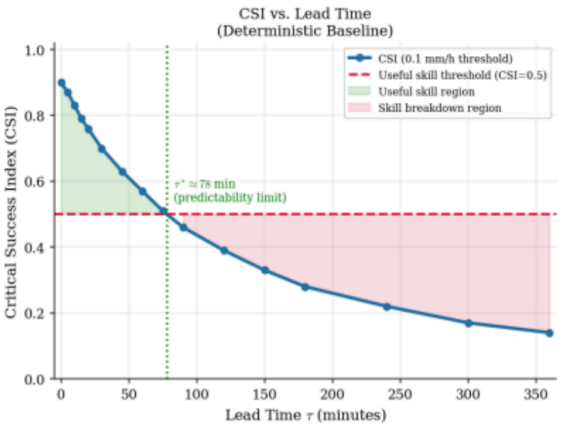

Exponential Spatial Correlation
$\rho(h) = e^{-h/\lambda}$
$\lambda = 10$ km
$\lambda = 20$ km
$\lambda = 40$ km
$1/e \approx 0.37$
Correlation $\rho(h)$
Distance h (km)

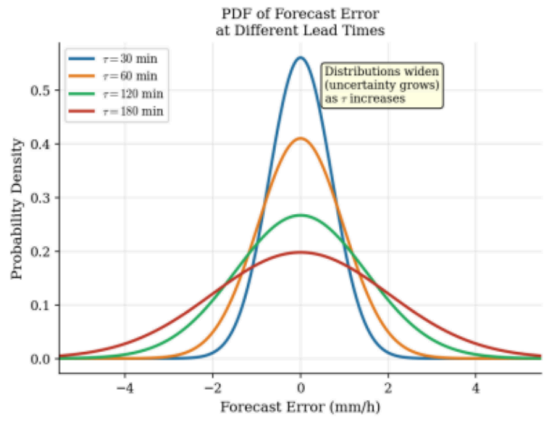

2D Spatial Correlation Field
($\lambda = 20$ km)
$\lambda = 20$ km
Δy (km)
Δx (km)
$\rho(h)$

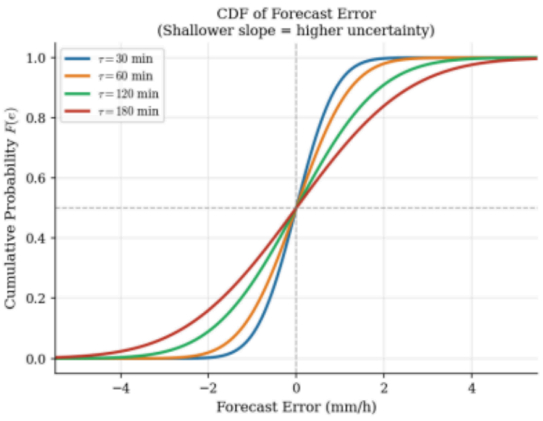

Observed $R(x,t)$ + Velocity $V(x,t)$
(Lucas-Kanade optical flow, navy arrows)
y (km)
x (km)
mm/h

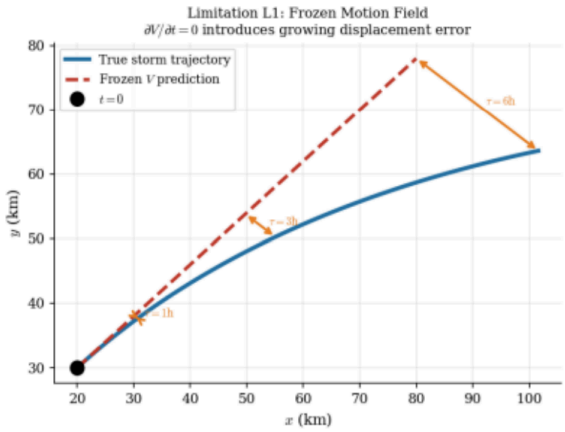

Forecast $\hat{R}(x, t+\Delta t)$
Semi-Lagrangian advection: $X(x, t+\Delta t) = X(x - V\Delta t, t)$
y (km)
x (km)
mm/h

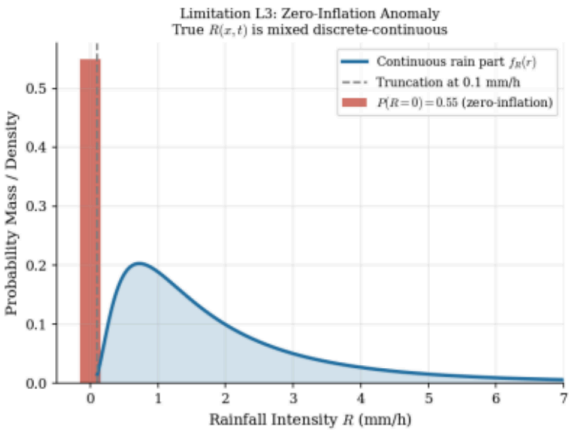

Forecast Error Growth with Lead Time
(Deterministic Lagrangian Persistence)
High accuracy
Moderate skill
Rapid degradation
RMSE
MAE
Error (mm/h)
Lead Time τ (minutes)


CSI vs. Lead Time
(Deterministic Baseline)
CSI (0.1 mm/h threshold)
Useful skill threshold (CSI=0.5)
Useful skill region
Skill breakdown region
$\tau^* \approx 78$ min
(predictability limit)
Critical Success Index (CSI)
Lead Time τ (minutes)


PDF of Forecast Error
at Different Lead Times
$\tau = 30$ min
$\tau = 60$ min
$\tau = 120$ min
$\tau = 180$ min
Distributions widen
(uncertainty grows)
as τ increases
Probability Density
Forecast Error (mm/h)


CDF of Forecast Error
(Shallower slope = higher uncertainty)
$\tau = 30$ min
$\tau = 60$ min
$\tau = 120$ min
$\tau = 180$ min
Cumulative Probability $F(e)$
Forecast Error (mm/h)


Limitation L1: Frozen Motion Field
$\partial V/\partial t = 0$ introduces growing displacement error
True storm trajectory
Frozen V prediction
$t = 0$
$\tau = 6$h
$\tau = 3$h
$\tau = 1$h
y (km)
x (km)


Limitation L3: Zero-Inflation Anomaly
True $R(x,t)$ is mixed discrete-continuous
Continuous rain part $f_R(r)$
Truncation at 0.1 mm/h
$P(R=0) = 0.55$ (zero-inflation)
Probability Mass / Density
Rainfall Intensity R (mm/h)